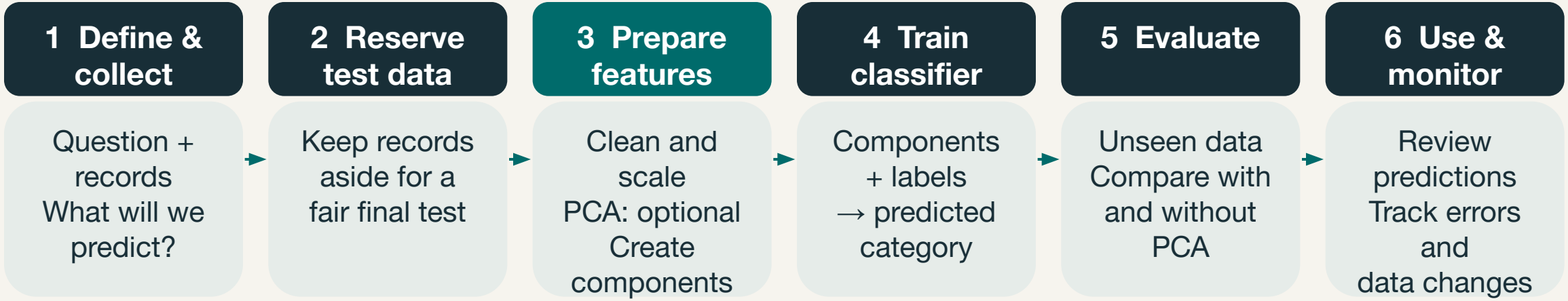
Principal Component Analysis

Making sense of public procurement data

Where PCA fits in a data-science pipeline

Feature engineering means preparing the numerical inputs a model will use.



Learn scaling and PCA from training data only; reuse them on test and new records.

PCA creates components without category labels. The classifier learns from labels.
PCA is not a classifier. It also works on its own for summaries and visual exploration.

Method example: scikit-learn, PCA → logistic regression

Train/test guidance: avoiding data leakage

2026 evidence: helping suppliers find tenders

The task

Classify tenders into procurement categories so suppliers can find relevant opportunities.

PCA prepares inputs

14 variables → 10 components
89.82% of variation retained

A classifier predicts

Random Forest: 91.17% accuracy on the study's selected categories. This is not PCA accuracy.

2024 data; 2026 paper. Research, not nationwide deployment.
The abstract does not establish that PCA outperforms the original variables.

Caroca et al. (2026), Machine Learning Classification of Public Tenders Using PCA — click for abstract

2025 evidence: summaries and smaller searches

Audit reports
Brazil · working paper

Dictionary counts + PCA → a summary index.
Explained 71–73% of variation in human-coded corruption counts in the validation samples.
Use: summarise reports; verify against human review.

Muço (2025), Measuring Corruption from Text Data

Document search
PCA-RAG · preprint

3,072 numerical features → 110 components.
Reported ~28.6× smaller indexes and up to 60× faster retrieval, with lower similarity quality.
Use: test a speed–quality trade-off on your documents.

Khaledian et al. (2025), PCA-RAG — click for preprint

Research results are specific to the data and evaluation used.

Six offices, six indicators

Office	Contracts per staff	Processing days	Amended contracts %	Average bids	Single-bid tenders %	Largest supplier %
A	41.3	35.2	11.1	5.3	17.3	27.3
B	31.3	26.5	10.2	6.4	17.5	29.3
C	68.2	46.2	20.5	6.0	21.6	22.8
D	69.5	51.6	19.2	3.5	35.0	35.1
E	50.0	34.1	12.6	3.3	30.6	35.7
F	56.3	41.7	15.4	5.5	30.4	30.2

In pairs: which offices look similar? What makes comparison difficult?

Synthetic classroom data: six of 60 invented offices. These are not Chilean records.

Some indicators move together

Workload pattern

More contracts per staff
Longer processing times
More amended contracts

Market pattern

Fewer bids
More single-bid tenders
A larger share for one supplier

PCA looks for shared patterns across numerical columns.

Teaching example: designed patterns in synthetic data. Method: scikit-learn PCA documentation.

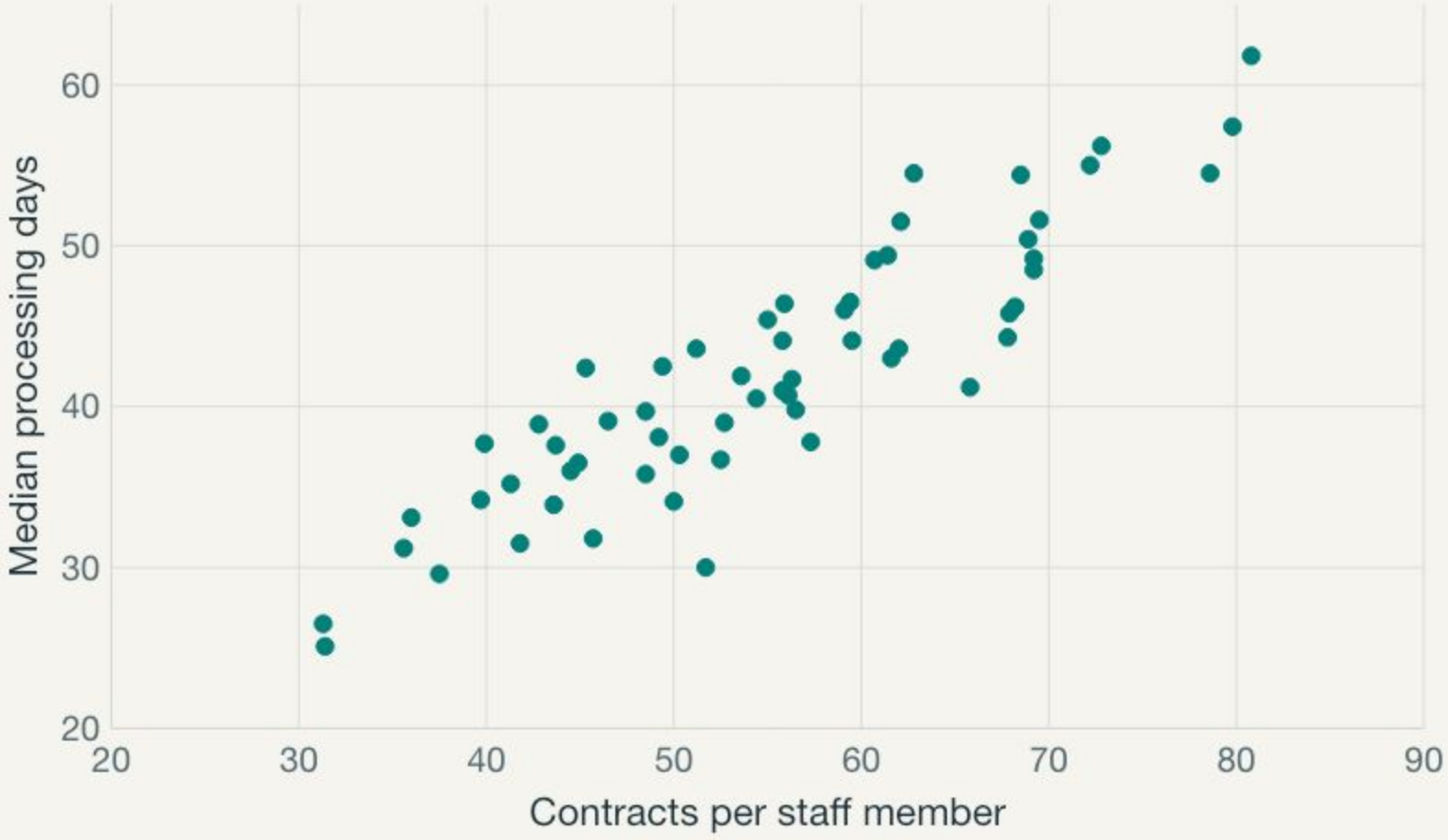
What PCA produces

Input	A table of offices and numerical indicators
Components	New summary measures that combine the original indicators
Scores	Each office’s position on those summary measures

PCA creates model inputs or an exploratory summary; it does not predict a class.

Method reference: scikit-learn, Principal Component Analysis (PCA). Click to inspect.

A useful viewing direction



One direction captures much of the shared pattern.

Each dot is one office.

Fictional data.

Source: synthetic classroom data. Method reference: scikit-learn PCA.

Units can change the result

The problem	Euros, days and percentages use different numerical scales.
Our choice	Put each indicator on a comparable scale before PCA.
What zero means	The average value for that indicator in this dataset.

Comparable scales give each column the same starting spread.

Method reference: scikit-learn, Importance of Feature Scaling. Click for a worked example.

A credible input table

Comparable work	Same reporting period and similar purchasing categories
Clear definitions	Agreed units, denominators and counting rules
Checked records	Missing values and unusual observations reviewed
Useful variables	Measurements that answer a real management question

Indicator definitions and data checks: Open Contracting Partnership, Red Flags guide (2024).

The meaning of each component

Indicator	PC1 weight	PC2 weight
Contracts per staff	0.58	0.01
Processing days	0.58	0.01
Amended contracts (%)	0.58	0.01
Average bids	0.01	-0.58
Single-bid tenders (%)	-0.01	0.58
Largest supplier share (%)	-0.01	0.58

PC1

Workload and processing pressure

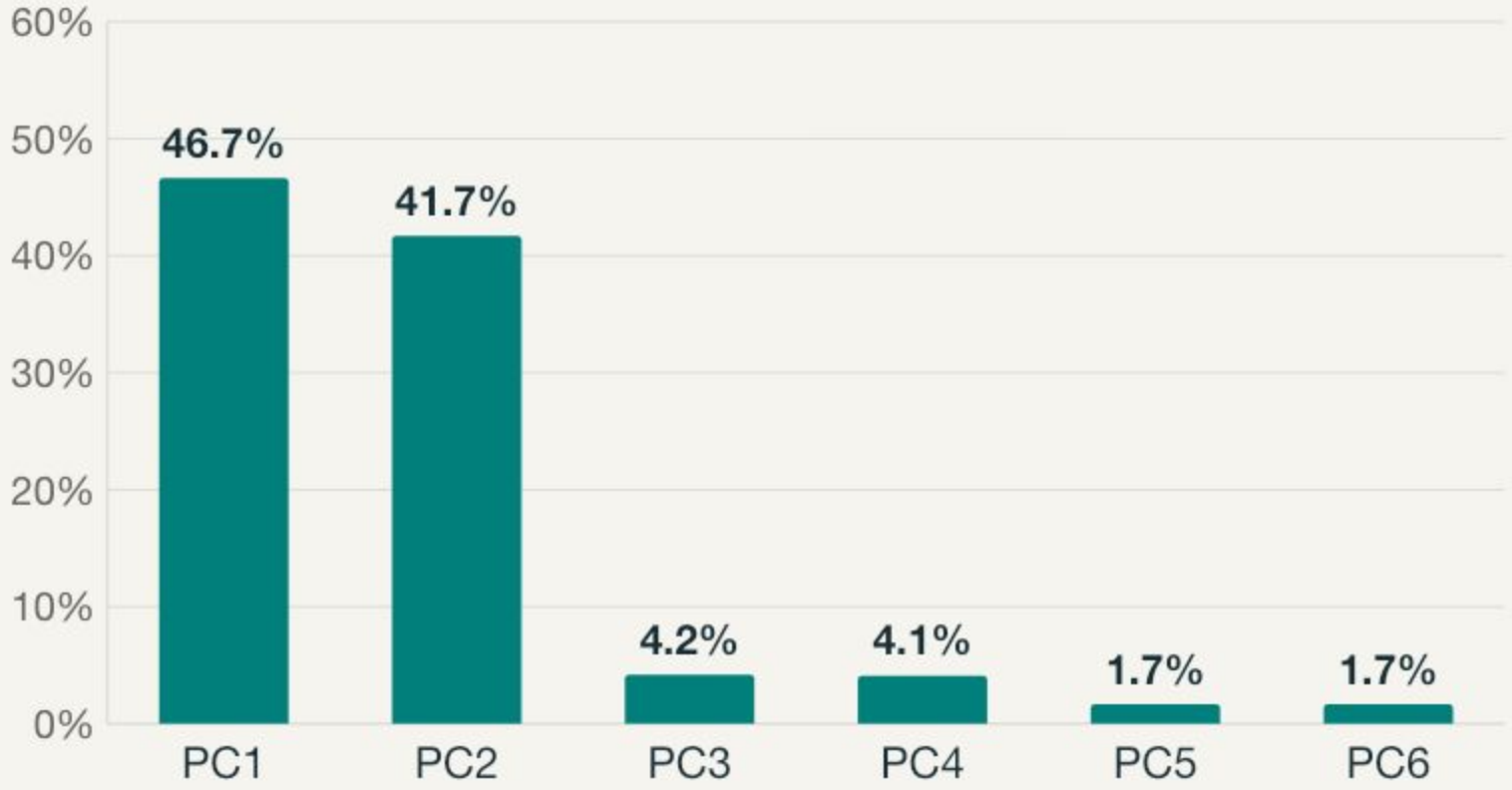
PC2

Supplier concentration

Calculated from fictional data. Labels describe patterns, not causes.

Method reference: scikit-learn PCA. Weights come from this lesson's synthetic data.

How much detail remains?



88.4%

of variation remains in the first two components.

11.6% remains outside the two-dimensional map.

Source: PCA calculation on 60 synthetic offices. 88.4% is a classroom result, not Chile's result.

A map of procurement profiles



Right

More workload and processing pressure

Up

Greater supplier concentration

Fictional data. Two components retain 88.4% of variation.

Pair activity: what would you check?



Choose C, D or E

1. What does the map show?

2. What might explain it?

3. What evidence would you check next?

Synthetic classroom data. 3 minutes in pairs, then 2 minutes to share.

Debrief: observation and follow-up

Office	Supported observation	Useful next check
C	Higher workload pressure, lower concentration	Staff capacity and case complexity
D	Higher on both components	Supplier market and procurement case files
E	Higher concentration, lower workload pressure	Specialist purchases and number of eligible suppliers

An explanation needs evidence beyond the map.

Illustrative interpretation of synthetic offices. No real organisation is assessed.

An unusual profile needs context

Office D looks unusual.

Possible explanations include specialist purchases, emergency needs, market concentration or errors in the records.

A defensible next step is a documented review of the underlying cases.

Risk interpretation: OCP Red Flags guide (2024). Office D is a fictional teaching example.

Procurement applications

Office comparison	Where do workload and processing patterns differ?
Supplier performance	Which suppliers share similar delivery and quality profiles?
Contract monitoring	Which combinations of delays, amendments and cost increases need review?

Proposed applications, not documented deployments. Indicator guidance: OCP (2024).

Other public-sector applications

Buildings: study	University of Girona: PCA-based monitoring of electricity consumption (2014).
Services: proposal	Waiting times, backlogs and repeat visits
Surveys: proposal	Related ratings of workload, support and communication

Published building case: Energy Procedia 62 (2014), 555–564. Other examples are proposals.

When PCA helps

PCA can help when…	A simpler approach may fit when…
Many numerical indicators overlap	One or two indicators answer the question
You need to explore similar profiles	You need a total, a trend or a specific rule check
You can interpret and check the resulting components	You need a decision based on explicit policy criteria

The management question determines the method.

Method reference: scikit-learn, Principal Component Analysis (PCA). Click to inspect.

Exit ticket

1. What does a principal component combine?
2. Does 88.4% retained variation mean 88.4% accuracy?
3. What should follow an unusual point on the map?
4. Which step predicts a category: PCA or the classifier?

What to take back to work

A component	Combines original numerical indicators into one summary
Retained variation	Describes how much of the data’s spread remains
An unusual point	Prompts a check of records and context
PCA’s role	Creates summary features; a classifier uses them to predict labels

Method reference: scikit-learn, Principal Component Analysis (PCA). Click to inspect.

2026 evidence: advanced uses of PCA

Compressing models
ICLR conference paper

Zhou et al. use robust PCA to separate model weights into low-rank and sparse parts.
Why it matters: PCA-related methods still support compression research.
Robust PCA is an extension of ordinary PCA.

Zhou et al. (2026), Large Language Model Compression with Global Rank and Sparsity Optimization

Inspecting numbers
AAAI student abstract

Davies et al. use PCA and linear models to examine how language models represent numbers.
They report noise and limits with decimal precision.
Why it matters: PCA can help inspect representations.

Davies et al. (2026), Language Models Do Not Embed Numbers Continuously — click for student abstract

Further reading: evidence of research use, not public-sector deployment.

Sources and what they support

Chile study	Caroca et al. (2026), LACCEI 14 variables, 10 components Click for the university abstract
PCA method	scikit-learn: PCA and feature scaling Components and retained variation Click for the method documentation
Procurement risks	OCP (2024), Red Flags guide Procurement indicator definitions Guidance, not proof of PCA deployment
Building case	Electrical consumption of academic buildings Energy Procedia 62 (2014), 555–564 Click for DOI: 10.1016/j.egypro.2014.12.417

Reference appendix. The main lesson remains 60 minutes. Sources checked 4 November 2026.

Responsible use of AI in this lesson

Trace the claim	AI helped research and draft the deck. Open the original source before repeating a claim.
Check the scope	Chile: find “14” and “10” in the abstract. Data: 2024. Publication: 2026. Check the sample and what the result means.
Show the provenance	Offices A–F are invented teaching data. 88.4% comes from our PCA calculation. Service and survey examples are proposals.
Keep a record	Record the source, date and exact claim. Distinguish abstract checks from full-paper review. Take responsibility for the final wording.

Reference appendix. The main lesson remains 60 minutes. Sources checked 4 November 2026.